

BADR ALDEEN AL-KHAZAN

Industrial Engineering, Operations Research & Decision Analytics, Supply Chain & Sustainable Systems

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PROFESSIONAL SUMMARY

Final-year CAPM-certified Industrial Engineering student (GPA 3.91/4.00) focused on applied Operations Research and Decision Analytics for supply chain, manufacturing, and sustainable systems. Experience spans simulation, optimization, queueing, stochastic modelling, predictive maintenance, and data-driven decision support, with active research and publication experience.

EDUCATION

B.Sc. Industrial Engineering (International Program), Universitas Islam Indonesia *Expected 2027*

GPA: 3.91/4.00. **Relevant coursework:** Operations Research, Production Planning, Simulation & Modelling, Facility Layout, Six Sigma Quality, Manufacturing Processes.

EXPERIENCE & LEADERSHIP

Simulation & Modelling Trainee, Delsim Laboratory, Universitas Islam Indonesia *Jul 2025 – Sep 2025*

- Built and validated discrete-event and system-dynamics models using FlexSim and PowerSim to analyze industrial system behavior.

Deputy Head, Events Division & Industrial Engineering Ambassador, UII Global (Volunteer) *Jul 2025 – Dec 2025*

- Led academic and cultural events, logistics and partnerships; served as PIC for international-student activities.

Deputy Head, Social Responsibility Circle, Yemeni Students' Union in Indonesia

- Coordinated student-support and community programs and major cultural events.

RESEARCH & PUBLICATIONS

Published	How Depending on AI Can Impact the Analytical Skills and Critical Thinking of Young People , <i>UII Repository</i> , 2026
Published	Germany's DIGITAL Cooperative Framework: Advancing Sustainable, Inclusive, and Innovative Global Digital Infrastructure , <i>UII Repository</i> , 2025
Accepted	Simulation-Based Analysis of the Bullwhip Effect in a Multi-Echelon Bottled Water Supply Chain , <i>Accepted for publication; in production</i> , 2026
Conference	Calibration and Cost-Sensitive Threshold Selection for Imbalanced Predictive Maintenance: an AI4I 2020 Study , <i>Accepted for presentation, TMIC 2026</i>
Under Review	Supply Chain Governance in Sustainability Reporting: a SCOR Orchestrate Analysis of an Indonesian State-Owned Enterprise , 2026
Under Review	Reducing University Motorcycle Exit Time through Automated Parking Access Control (SIGAP) , 2026
Under Review	Teaching Sustainability with Unsustainable Tools: Toward a Reflexive Green-AI Literacy Framework for Education for Sustainable Development , 2026
In Progress	Green Vehicle Routing for Electric Truck Distribution with Load-Dependent Energy Consumption , 2026

AWARDS & CERTIFICATIONS

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| <ul style="list-style-type: none">1st Place: EXPO PSIT 2026, SIGAP RFID Automated Parking Access SystemBest Presentation: EXPO RSKE 2025, Multifunctional Ergonomic PenFuture Global Leaders Scholarship: Universitas Islam Indonesia (2023–2027)TOEFL iBT 90/120: ETS (2026–2028) | <ul style="list-style-type: none">1st Place: Idea Champion 2.0, AI-PHL GuardianOutstanding Delegate: IYEC 12, Malaysia & SingaporeCAPM®: Project Management Institute (2026–2029)Aspire Leaders Program: Aspire Institute (2025) |
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SKILLS & LANGUAGES

Technical: FlexSim, PowerSim, simulation, queueing analysis, stochastic modelling, supply chain analytics, SPC & Six Sigma, production planning, facility layout, work measurement, database design, Advanced Excel, technical writing.

Professional: Project management, analytical thinking, team leadership, cross-cultural communication, event management, public speaking.

Languages: Arabic (Native), English (B2; TOEFL iBT 90), Indonesian (B2), German (A1).